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Analysis I - Chapter 3

3.1 - Fundamentals

1. Let $A = \{a, b\}$ and $B = \{c, d\}$. Since $\{a, b\} = \{c, d\}$, $A = B$. By A3.2, we have all elements of A being elements of B; and vice versa. Additionally, since A & B are non-empty, by L3.1.5 we have an object x such that $x_1 \in A$ and $x_2 \in B$. Thus:
$a \in A \Rightarrow a \in B$. Assign this arbitrarily to c.
$\Rightarrow a = c$ therefore implies $b \in A \Rightarrow b \in B$ and since c already belongs to A, $b = d$. A similar proof is clear if we choose $a = d$.

2. By A3.1 we have all of these sets being objects. Therefore though $\emptyset$ is a set, there is no element x such that $x \in \emptyset$ by A3.3. Therefore, by L3.2, none of the three other sets are equal to $\emptyset$, unless it is precisely $\emptyset$. By 3.1 we have $\{\emptyset\}$ being an object as well. By 3.4, the singleton set will not be equal to any other as its sole element is $\emptyset$. Therefore, $\{\{\emptyset\}\}$ follows similar logic as its single element is $\{\emptyset\}$. With $\{\emptyset, \{\emptyset\}\}$, we already know it must be distinct given that it is not a subset of any of the previous sets. Therefore there are some elements in $\{\emptyset, \{\emptyset\}\}$ that are not found in any other set by themselves.

3. We prove $A \cup A$. By A3.5 we have $x \in A$ or $x \in A$, which regardless of the element x will hold true for both. Therefore by A3.2 since all elements of A are also elements of A, we have all elements of A found in $A \cup A$ with no element not found in A. $\therefore A \cup A = A$. For the commutative operation we use A3.5's inclusivity. Since either $x \in A$ or $x \in B$ or both must be true to be included in $A \cup B$, we have $A \cup B \Leftrightarrow (x \in A$ or $x \in B) = B \cup A \Leftrightarrow (x \in B$ or $x \in A)$ being identical as any element x is still being checked for the same 2 conditions $x \in A$ or $x \in B$. Therefore $A \cup B = B \cup A$. Thus we have $A \cup \emptyset = \emptyset \cup A$. Now since there is no element $x \in \emptyset$ by A3.3, we still have
$A \cup \emptyset \Leftrightarrow (x \in A$ or $x \in \emptyset)$ (by A3.5)
$\Rightarrow A \cup \emptyset \Leftrightarrow (x \in A)$. Therefore since all elements of A satisfy $x \in A$ by D3.1.1 & L3.1.5, then $A \cup \emptyset = \emptyset \cup A = A \cup A = A$.

4. We first start with $A \subseteq B$ and $B \subseteq A$ implying $A = B$. Since for any object x, $A \subseteq B \Rightarrow x \in A$ and $x \in B$. Similarly, $B \subseteq A \Rightarrow x \in B$ and $x \in A$. Therefore all elements of A are elements of B, and vice versa. Therefore by A3.2 we have $A = B$. For $A \subsetneq B$ and $B \subsetneq C \Rightarrow A \subseteq C$, we pick some arbitrary element of A. We have $x \in A$ and $x \in B$, but B contains and L3.1.5, then elements which $x \notin A$. Then, since $B \subsetneq C$, all elements $x \in A$ and $x \in B$ also have $x \in C$, though there are elements in C not in B, and since all elements in B $x \in B$ and $x \in C$, we have some elements $x \in B$ and $x \in C$ but, as demonstrated, $x \notin A$. $\therefore A \subsetneq C$.

5. If $A \subseteq B$, we have all elements of A $x \in A$ and $x \in B$. Therefore $A \cup B = B$ implies that all elements of A are contained in B, as for all $x \in A$ we create a set that is equal to B. $\therefore A$ is a subset of B. Now, $A \cap B = A$ implies that all elements of A are also found in B, since $A \cap B \Leftrightarrow x \in A$ and $x \in B$ for any arbitrary element. Therefore if an element x has $x \in B$ but $x \notin A$, then it is not in $A \cap B$. But as discussed we have all $x \in A$ also $x \in B$. therefore A is a subset of B by D3.1.14

7. $A \cap B$ implies only elements x that $x \in A$ and $x \in B$ are kept. Since all elements in $A \cap B$ must $x \in A$, it follows $A \cap B \subseteq A$. Similarly, since all elements in $A \cap B$ must $x \in B$, it follows $A \cap B \subseteq B$ by D3.1.14. If $C \subseteq A$ and $C \subseteq B$, this implies that all elements in C are found in A and B. This is analogous to $C \subseteq A \cap B$, as it must follow that for $C \subseteq A \cap B$, an element x in C $x \in A$ and $x \in B$ by definition. Thus $A \subseteq A \cup B$ as all elements in A are by definition found in $A \cup B$ - which must have the same (or more) elements than A. This applies for $B \subseteq A \cup B$ as well. If $A \subseteq C$ and $B \subseteq C$, this means all elements in A or B are found in C; if $x \in A$ or $x \in B$, then $x \in C$. Notice that this is analogous to A3.5.

$\Rightarrow A \cup B \subseteq C$.

6 a. $A \cup \emptyset = A$. By L3.1.12. $A \cap \emptyset = \emptyset$ as $A \cap \emptyset$ implies that for an element x, $A \cap \emptyset \Rightarrow x \in A$ and $x \in \emptyset$. By A3-3, $x \notin \emptyset$ for all x. Therefore no element is in $A \cap \emptyset \Rightarrow A \cap \emptyset = \emptyset$.

b. from E3.15, we have already shown that if $A \cup X = X$, this implies $A \cap X = A$

c. If we have $A \cap A$, then all elements in A must also be found in A such that it is in $A \cap A$. Thus since $x \in A$ and $x \in A$ applies for all x in A, $A \cap A = A$. As for $A \cup A$, any element $x \in A$ is also $x \in A \cup A$. No other element outside from these are included $\Rightarrow A \cup A = A$

d. from L3.1.12 we have already stated $A \cup B = B \cup A$. Similarly, $A \cap B$ holds that each of its elements has $x \in A$ and $x \in B$. This is the same as $x \in B$ and $x \in A$ as the ordering is irrelevant in D.3.1.1

e. from L3.1.12 this has already been proven. We hold $(A \cap B) \cap C$ to also hold the axiom of substitution, thus the proof is analogous to $(A \cup B) \cup C = A \cup (B \cup C)$.

f. Take one element x in $x \in B \cup C$. Therefore any element in B or C must be found in A to satisfy $A \cap (B \cup C)$. Similarly, Any element found in A and B that writes with all elements found in A and C satisfy $(A \cap B) \cup (A \cap C)$. Similar logic is applied to $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

g. $A \cup (X \setminus A)$ implies all elements of A in X are removed $\Rightarrow X \setminus A$. They are thus re-introduced with $A \cup (X \setminus A)$, therefore cancelling out the operation $\Rightarrow A \cup (X \setminus A) = X$. On the contrary, $A \cap (X \setminus A)$ takes all elements $x \in A$ in X. Thus, there will be no element in $A \cap (X \setminus A)$ such that $x \in A$ and $x \in X$ therefore we have $A \cap (X \setminus A) = \emptyset$

h. $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$ as the action is simply removing all elements $x \in A$ or $x \in B$ in X. In a similar fashion, $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$ removes all elements $x \in A$ and $x \in B$ in X.

8. $A \cap (A \cup B) = A$

We first identify $A \cup B$. All elements of both $x \in A$ or $x \in B$ are in $A \cup B$. Thus, $A \cap (A \cup B)$ takes all elements such that:

$x \in A$ and $(x \in A$ or $x \in B)$.

Thus, only elements that are found in A will be included per D. 3.1.22

$\therefore A \cap (A \cup B) = A$. Similarly, $A \cup (A \cap B)$ first eliminates, all non-A elements — some (or all) elements are kept in $A \cap B$ — no element x such that $x \notin A$. Thus $A \cup (A \cap B)$ joins all other elements in A previously eliminated by the intersection of $\cap B$.

$\therefore A \cup (A \cap B) = A$.

9. We have $A \cup B = X$ and $A \cap B = \emptyset$. It must therefore follow

$A = X / B \Rightarrow A = (A \cup B) / B$

We have some elements x such that $x \in A$ or $x \in B$ in $A \cup B$. It therefore follows that all elements $x \in B$ are removed in $(A \cup B) / B$. Thus only $x \in A$ is left; equivalent to A. Similarly we have

$B = X / A \Rightarrow B = (A \cup B) / A$

Therefore since all elements $x \in A$ and $x \in B$ are kept, we thus have all elements $x \in A$ removed from $(A \cup B) / A$. Thus only elements $x \in B$ are left;

$B = X / A$.

10. $A \setminus B$ removes any elements in A such that $x \in B$.

$A \cap B$ removes any elements in A such that $x \notin B$ or in B such that $x \notin A$. Thus, by definition, $A \setminus B$ removes all elements such that $A \cap B$ keeps. Similarly, $B \setminus A$ keeps elements unique to B — not A. Thus differing from $A \setminus B$ and by definition also differing from $A \cap B$.

11. A3.7 states that we can replace some variable z with a variable y that satisfies some property $P(x, y)$ for some $x \in A$. Thus we say that $P(x, y)$ is not dependent on y — $P(x, y)$ is true regardless of y and depends only on x to give $P(x)$. Then, the set $\{x \in A : P(x)\}$ contains only the elements x such that $P(x)$ is true. If we have $z \in \{x \in A : P(x) \text{ is true}\}$ this implies $z \in \{x \in A : P(x, y) \text{ is true}\}$, with y we defined $P(x, y)$. Thus A3.7 implies A3.6

12. i. $A' \cup B'$ will give all elements $x \in A'$ or $x \in B'$. Thus, by D3.1.14 we have all elements in $x \in (A' \cup B')$ in either A or B. Thus, $A \cup B$ represents this union and we have $A' \cup B' \subseteq A \cup B$ as there may be some elements in $(A$ or $B)$ not found in $(A'$ or $B')$

ii. $A' = \{1, 2, 3\}$ $B' = \{4, 5, 6\}$

$A = \{1, 2, 3, 4, 5, 6\}$ $B = \{1, 2, 3, 4, 5, 6\}$

We thus have $A' \subseteq A$ and $B' \subseteq B$ but, $A' \setminus B' = \{1, 2, 3\}$ and $A \setminus B = \emptyset$. By D3.1.14, $\{1, 2, 3\} \notin \emptyset$.

A true statement would be:

$(A' \cup B') \setminus B' \subseteq (A \cup B) \setminus B$. $(A' \cup B') \setminus B'$ gives us all elements in A' or B'. Then all elements $x \in B'$ are removed $(A' \cup B') \setminus B' =$

Similarly, $(A \cup B) \setminus B = A$. Thus, $(A' \cup B') \setminus B' \subseteq (A \cup B) \setminus B$

13. $A = \{x_1\}$. Thus $A' = \emptyset$ for $A' \subsetneq A$ to hold as having any $A' = \{x_2\}$ with $x_1 \neq x_2$ goes against D.3.1.14 as $x_2 \notin A$. Conversely, if $x_2 = x_1$ we have $A = \{x_1\} = A' = \{x_1\}$. Thus $A = A'$, and by D.3.1.14, A' is not a proper subset of A.

3.2 - Russell's Paradox (Optional)

1. We get A3.3 by defining some property $P(x)$ such that $x \notin A$. Therefore we have $A = \{x : x \notin A\} = \emptyset$. there is the element satisfying this.

A3.4 can also be generalized by saying that $P(x)$ is some property whereby $A = \{x : P(x) = c_1\}$ where c is some object. We thus also get $A = \{x : P(x) = a \text{ or } P(x) = b\}$ for 2 separate constants a, b. Similarly, we create $A \cup B = \{x : x \in A \text{ or } x \in B\}$ whereby $P(x)$ is this property $x \in A$ or $x \in B$. A3.6 and A3.7 are close revisions of A3.4, whereby this property $P(x)$ is modified for some $x \in A$ (A3.6) or for two variables, $x \in (A, B)$ (A3.7).

3. By A3.4, we can postulate a set Ω such that regardless of the object x, $P(x)$ will always be true. Therefore, only and all objects are included in $\Omega = \{x : P(x) \text{ is true}\}$

3.3 - Functions

1. We have $f: X \to Y$ for the substitution and $f: X \to Y$. Since property, if this $X = X$, $Y = Y$, and $f \& f$ follow that iff agree for all $x \in X$, f follows $f = f$. Then by P3.3.8 we already have $f = g$ iff $X = X'$, $Y = Y'$ and $f(x) = g(x)$ for all $x \in X$. Then, suppose $h: X'' \to Y''$ whereby $X'' = X'$ and $Y'' = Y$. If $h(x) = g(x)$ for all $x \in X$, then $h = g$ by definition, and it follows $h = g = f$ $\Rightarrow h = f$.

2. We first begin with $A = \{c\}$ for some constant c. It thus follows that since c is a primitive object, not a set, $A \notin A$. Additionally, by A3.10, we have that there must be one disjoint element from A. Ergo, this implies that A must infinitely contain itself for $A \in A \Rightarrow$ eventually we boil it down to some primitive object c. And from $A = \{c\}$, it cannot contain itself. We know $A \notin B$ or $B \notin A$ given that there must be a hierarchical system in either A or B such that if A is contained in B, B must therefore itself be an object one "degree" higher than B. This reduces around the singleton Axiom, whereby $A = \{c\}$ indicates $A \notin a$. The opposite can be easily verified.

2. For $g \circ f$, it therefore follows that $g \circ f : X \to Z$. Thus, if both f and g are injective then each $y \in Y$ is carved out by a distinct $x \in X$. Similarly, each $z \in Z$ is carved out by a distinct $y \in Y$. Thus, each $x \in X$ maps to a $y \in Y$, which then maps to some $z \in Z$. thus $g \circ f$ is also injective. Similarly, if each and all $y \in Y$ are derived from some $x \in X$, and each and all $z \in Z$ are derived from some $y \in Y$, then the following follows: Each $z \in Z$ is derived by some $y \in Y$, whereby each $y \in Y$ is derived by some $x \in X$. Thus each $z \in Z$ is derived by some $x \in X$. Thus for some $z \in Z$, there is an x $f(x) = z$. Thus $g(f(x))$ is surjective.

3. Suppose we have an empty set $\emptyset$. By A3.3, we have no x such that $x \in \emptyset$. Therefore there cannot be any x & x' such that $x = x'$ but $f(x) = f(x')$. Thus $f: \emptyset \to X$ is vacuously injective for all X. However, surjectivity requires that $x \in \emptyset$, which is absurd. Thereby from having no $y \in X$, we have $f: \emptyset \to X$ iff $X = \emptyset$, vacuously. Finally, a function $f: \emptyset \to X$ is bijective at the intersection of properties injectivity & surjectivity. Therefore $f: \emptyset \to X$ is bijective iff $X = \emptyset$. QED

5. $g \circ f$ being injective means that the input-output relation is unique for all $x \in X$. If f were not injective, some $x \in X$ & $x' \in X$ result in the same $y \in Y$, thereby leading to a $z \in Z$. Thus, though we have the same outputs y leading to z, these may have come from different inputs and as such different domains. But since $g \circ f$ is injective, both the domain and codomain must be unique. Therefore, it is a contradiction. g doesn't need to be injective so long it produces the same z. $g \circ f$ correlates $g \circ f = X \to Z$. The intermediate Y set is negligible.

If f^{-1} is invertible and has f as its inverse, we simply rely on the wordplay of the f verbatim. Since f is the inverse of f^{-1}, the inverse of f^{-1}, denoted as $(f^{-1})^{-1}$ is f. We therefore have $(f^{-1})^{-1} = f$.

QED. This is merely diction being transformed into notation.

4. If we have $g \circ f = g \circ \tilde{f}$ with f and $\tilde{f}$ sharing domain & codomain, it follows that $f = \tilde{f}$ as by D3.13 we define $g \circ f: X \to Z$. Thereby if $g \circ \tilde{f}$ is equal for all elements in X to Y to Z, then by B.3.5 they are equal. It should be noted that injectivity is required. Since f and $\tilde{f}$ share both domain and codomain, applying a function g to this codomain regardless results in Z. Thus, a counterexample could imply f and $\tilde{f}$ sharing X and Y, but applying different rules to these sets that are "standardized" by g.

Now, let $g \circ f = \tilde{g} \circ f$ with f being surjective, then all $y \in Y$ have some $f(x) = y$. Therefore the same codomain Y is outputted, which is thereby transformed by g or $\tilde{g}$ to produce the output z. Surjectivity is not required as the function $f: X \to Y$ maps any $x \in X$ to a $y \in Y$, regardless of which $y \in Y$ are seen or not. The importance lies in that it is the same transformation done by f, which x or x' then goes into $g \circ f$ or $\tilde{g} \circ f$.

6. Let a sample $x \in X$ be transformed by f. Since f is bijective, we result with some unique $y \in Y$. Assuming f^{-1} is also bijective, this $y \in Y$ is then mapped back to the original $x \in X$ as there is one and only one element x where $f(x) = y$. Similarly, a unique $y \in Y$ is mapped to a unique $x \in X$ by $f(y)$ with no overlap. By applying its inverse this unique $x \in X$ returns back to its unique one-to-one correspondent $y \in Y$. We therefore have $f^{-1}(f(x)): X \to Y \to X$ and since f^{-1} and f are both bijective, $f^{-1}(f(x)) = x$.

* identity maps are the equivalent of 1 in set theory!

7. Let $g \circ f : X \to Z$. Since both g and f are bijective we have all $z \in Z$ correspond to a unique $x \in X$ since $f: X \to Y$ maps each $x \in X$ to all and a unique $y \in Y$. Then, g does the same process with $g: Y \to Z$. Since all $z \in Z$ have some $g(f(x)) = z$ and $x = x'$ iff $g(f(x)) = g(f(x'))$, $g(f(x))$ is therefore bijective. Taking its inverse, we want a function whereby $Z \to X$. Thus, we go $Z \to Y$ through the inverse of g. Recall that both g and f are bijective, so our correspondence is unique and complete. Thus far, we have $g^{-1}(z) = y$. Thus, for all $y \in Y$ to correlate to a unique x, we have the inverse of f map these two sets. Therefore we have:

$f^{-1}(y) = x$, or $f^{-1}(g^{-1}(z)) = x$,

which is therefore $f^{-1} \circ g^{-1}$ and is equivocally $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

c. Given $f: A \to B$ being bijective, $f \circ f^{-1}$ first maps all $f^{-1}: B \to A$ uniquely and completely. For any and every $b \in B$, there is a unique $a \in A$. this $a \in A$ is then mapped uniquely and completely to an element $b \in B$, which by previous examples, must give $b = b'$. Therefore there is no difference between the domain and codomain of $f \circ f^{-1}$ by A3.2. Thus, $\iota_{B \to B}$ relates all elements $b \in B$ to some $b' \in B$, which $b = b'$ by definition. Thus, $f \circ f^{-1} = \iota_{B \to B}$. The similar occurs with $f^{-1} \circ f = \iota_{A \to A}$, whereby any and all $a \in A$ are mapped back to some $a' \in A$, whereby $a = a'$. Thus, this is equivocal to the identity map on A, denoted $\iota_{A \to A}$. QED

various proofs relate to $f \circ f^{-1} = \iota_{B \to B}$

8. a If $X \subseteq Y \subseteq Z$, then $\iota_{X \to Y}$ maps all $x \in X$ to a $y \in Y$ such that $x = y$. We are therefore left with the same codomain X by A3.2 Then, $\iota_{Y \to ?}$ is really mapping all elements in X, which by $X \subseteq Y$ must also be in Y, which must therefore be in Z. By P.3.1.13, we have $X \subseteq Z$ and $\iota_{X \to Z} = \iota_{Y \to Z} \circ \iota_{X \to Y}$

b. Given $f: A \to B$, $f \circ \iota_{A \to A}$ maps all elements $a \in A$ back to itself. Therefore a is transformed to a, and there is no change in the domain. Since $\iota_{A \to A}$ maps a to the same elements and $A = A$ by A3.2, we can cancel this function and are left with $f = f$, D.3.3.8 trivially proves this. Similarly, $\iota_{B \to B}$ maps an element $b \in B$ to itself, and this correspondence leaves us with the same domain. Therefore this function is negligible, and we have: $f = \iota_{B \to B} \circ f \Rightarrow f = f$, which is trivially proven by D.3.3.8

d. With $h: X \cup Y \to Z$, $h \circ \iota_{X \to X \cup Y}$ maps all $x \in X$ to their correspondent $a \in X \cup Y$ such that $x = a$ for all $x \in X$. Therefore the domain X is equal to the codomain. Thereby we have X being mapped to Z via $h: X \cup Y \to Z$, where $X = X \cup Y$ in this scenario. Thus, assuming h maps these elements exactly like f, then $h \circ \iota_{X \to X \cup Y} = f$. The similar process is done for $h \circ \iota_{Y \to X \cup Y}$, although the codomain $X \cup Y$ is now equivalent to Y. Therefore $Y \to Z \Rightarrow h \circ \iota_{Y \to X \cup Y} = g$.

∴ If $f(x) = g(x)$ for all $x \in X \cup Y$, then the codomain produced by f and g are equivalent. Thus, regardless if $f \cap g = \emptyset$, we have some function $h \circ \iota_{X \to X \cup Y} = f$ and $h \circ \iota_{Y \to X \cup Y} = g$ whereby the elements mapped are unique to either f or g.

QED.

5. If f is surjective, then for any $s \in S$ $f^{-1}(s)$ is also defined. Thus, each $x \in f^{-1}(S)$, regardless if f is injective, goes back to S through the found image $f(x) \in f(f^{-1}(S))$. Thus, if f were not surjective, some elements $f^{-1}(S)$ are undefined, and therefore x ~~$\in$~~ s for some $s \in S$. Similarly, $f^{-1}(f(S))$ means that for some $s \in S$, the found image must be unique. Thus only this $s \in S$ produces $f(s) \in f(S)$. thereby the backwards image is also unique. $f^{-1}(f(s))$ produces s as only one element s is the backwards image of $f(s)$.

7. Each $f: X' \to Y'$ must be have $f \in Y^X$ by A3.11. We therefore have $Z := \{f : X' \subseteq X \text{ and } Y' \subseteq Y\}$. We treat this as $z \in Z \Rightarrow z$ maps each $x' \in X'$ to some $y' \in Y$ completely. By A3.7 this must exist, and therefore satisfies the criterion for a set.

10. Suppose $(\bigcup_{\alpha \in I} A_\alpha) \cup (\bigcup_{\alpha \in J} A_\alpha) \neq \bigcup_{\alpha \in I \cup J} A_\alpha$. Then there is some $x \in (\bigcup_{\alpha \in I} A_\alpha) \cup (\bigcup_{\alpha \in J} A_\alpha)$ such that $x \notin \bigcup_{\alpha \in I \cup J} A_\alpha$ and vice versa. x is found in either all elements of A_α $\alpha \in I$, A_α $\alpha \in J$, or both. And yet, x is not found in the collection of all elements of A_α such that $\alpha \in I$, $\alpha \in J$, or both. We therefore have a contradiction. For brevity, we will omit the second portion as it is the same exact problem reworded. I am getting a bit hurried.

6. I don't think I get the hints. Suppose we have a set $\{0,1\}^X$. Then we have $Z := \{\emptyset, \{0\}, \{1\}, \{0,1\}\}$. Each of these $z \in Z$ can be traced back to $X := \{4,7\}$ by some function f. More generally, each function f can map some $x \in X$ to give all elements for any given $z \in Z$. Thus, each $y \in Y$ has some configuration — defined by f — that rearranges elements in x to produce a mapping in Y. By definition, Y must be at most the same cardinality as X. Not one element $x \in X$ produces more than one $y \in Y$. Therefore each $z \in Z$ is defined to be derived from X in some surjective function.

8. Given any two sets A, B, there exists a set $A \cup B$, called the union of A and B, which consists of all $x \in \cup\{A, B\}$ by 3.12. We treat A and B as objects by 3.1 to formulate the pair set $\{A, B\}$. Thus, $x \in \cup\{A, B\} \iff (x \in S \text{ for some } S \in A)$, whereby $S = A$, $S = B$, or both by 3.4. The set of all these x denotes the set $A \cup B$.

9. Suppose for contradiction, $\{x \in A_\beta : x \in A_\alpha \ \forall \alpha \in I\} \neq \{x \in A_{\beta'} : x \in A_\alpha \ \forall \alpha \in I\}$. Then there is a $x \in A_\beta$ such that it is found in all $\alpha \in I$, but not in $A_{\beta'}$. However, $\beta' \in I$, and therefore invalidates $x \in A_\alpha \ \forall \alpha \in I$ for some $\alpha = \beta'$. We therefore have a contradiction, and our proof is done. $\square$. Thus, (3.4) is true as by A3.7 we have $y \in \bigcap_{\alpha \in I} A_\alpha$ imply $y \in \{x \in A_\beta : x \in A_\alpha \ \forall \alpha \in I\}$ $\Rightarrow \{y \in A_\beta : y \in A_\alpha \ \forall \alpha \in I\} \Rightarrow (y \in A_\beta \ \forall \alpha \in I)$ is true.

11. Take a sample $x \in X \setminus \bigcup_{\alpha \in I} A_\alpha$. This element x is found nowhere in $\bigcup_{\alpha \in I} A_\alpha$ but found in X. We thus replicate the action of taking away all $x' \in A_\alpha \ \forall \alpha \in I$. By definition, this is done with $\bigcup_{\alpha \in I} (X \setminus A_\alpha)$ as all sets containing these x' are removed. Then, $X \setminus A_\alpha$ is expanded to include these x. Similar logic applies to the second portion.

3.5 Cartesian Products

1. i. $(x,y) := \{\{x\}, \{x,y\}\}$ is a set from A3.4, whereby $a = \{x\}$ and $b = \{x,y\}$ forms the pair set $\{a,b\}$. $y \in \{a,b\}$ iff $y = a$ or $y = b$. By A3.1 b is an object. Thus, (x,y) forms a valid set and is an ordered pair. By definition, it follows (3.5).

ii. $(x,y) := \{x, \{x,y\}\}$ By A3.10 we have x being a primitive object. Additionally, because of E3.2.2, we know that primitive objects cannot be self-containing. Thus (x,y) forms an ordered pair and (3.5) is valid.

iii. By A3.7, $\{(x,y) : y \in A\}$ is a set as each $x \in y$ within the set satisfy the (x,y) property. As each (x,y) object is unique. Thus, $\{(x,y) : y \in A\}$ is a set and A3.12 ensures that if pairings of sets occur, they can still be broken down.

2. $(x_i)_{i \in n} = (y_i)_{i \in n}$ by D3.5.6, and therefore having $(X_i)_{i \in n}$ has the cartesian product be a set via the partial function in E3.4.7. Thus since all of these map to some arbitrary X, A3.6 is correct and therefore the cartesian product is also a set.

3. Given the ordered pair (x,y) and ordered n-tuple (x',y'), we have:

Reflexivity: $(x,y) = (x,y)$ by (3.5) as $x = x$ and $y = y$. Thus, both components are equal.

Similarly, $(x',y') = (x',y')$ holds by D3.5.6 as $\forall i \in n$, we have $x_i = y_i$. For example, at $i = 1$; $x' = x$ and at $i = 2$ $y' = y'$.

Symmetry: Suppose we have two ordered pairs (x,y) and (x',y'). Then if $(x,y) = (x',y')$, $(x',y') = (x,y)$ as we can replace x' and y with x and y given that each element is equivalent to eachother. Since $x' = x$ implies $x = x'$, $(x',y') = (x,y) = (x',y')$. A similar proof holds for n-tuples.

Transitivity: Let $(x,y) = (x',y')$ and $(x',y') = (x_1,y_1)$. By D3.5.1 $x = x'$ and $x' = x_1$, therefore $x = x_1$. Similarly, $y = y'$ and $y' \in y_1$, therefore $y = y_1$. We thereby have all elements (x,y) equal all elements (x_1,y_1), which proves $\{(x,y) = (x_1,y_1)\}$ by D3.5.1. This can be proved similarly with n-tuples.

4. $A \times (B \cup C)$ is a set of all pairs (a,d) whereby $a \in A$, $d \in B \cup C$ by D3.5.6. Then $(A \times B) \cup (A \times C)$ has $\{(a,b)\} \cup \{(a,c)\}$ for all $a \in A$, $b \in B$, $c \in C$. Joining these contains $\{(a,b),(a,c)\}$. Thus (x,y) is an element of this set iff $x \in A$ and ($y \in B$ or $y \in C \Rightarrow y \in B \cup C$). Therefore by A3.7 we have a correspondence, whereby each element is replaced by an element of $A \times (B \cup C)$ thus, since all elements are equivalent these sets must be equivalent to eachother. The remaining exercises are trivial parallels of this same notation.

$A \times (B \cap C) = (A \times B) \cap (A \times C)$

$\Rightarrow \{(a,d) : a \in A, d \in B \cap C\}$

$= \{(a,b) : a \in A, b \in B\} \cap \{(a,c) : a \in A, c \in C\}$

↳ Implies that $(x,y) \in \{(a,b) \dots c \in C\}$ iff $x = a$ $\exists a \in A$ and ($y \in B$ and $y \in C$) $\Rightarrow y \in B \cap C$.

5. $(A \times C) \cap (C \times D) \Rightarrow \{(a,b) : a \in A\ b \in B\} \cap \{(c,d) : c \in C, d \in D\}$

$\Rightarrow \{(x,y) : x \in A, x \in C, y \in B\ y \in D\}$

$\Rightarrow \{(x,y) : x \in A \cap C, y \in B \cap D\}$

Conversely:

$(A \cap C) \times (B \cap D)$

$\Rightarrow \{(x',y') : x' \in A \cap C, y \in B \cap D\}$

Therefore since they hold the same property $P(x,y)$, we can replace any x by x' and any y by y'. Therefore $\forall (x,y)$ there is a unique (x',y'). Thus,

the sets must be equivalent.

Now we have

$(A \times B) \cup (C \times D)$

$\Rightarrow \{(a,b) : a \in A, b \in B\} \cup \{(c,d) : c \in C, d \in D\}$

$\Rightarrow \{(x,y) : x \in A, y \in B, x \in C, y \in D\}$

$\Rightarrow \{(x,y) : x \in A \cup C, y \in B \cup D\}$.

Conversely, $(A \cup C) \times (B \cup D)$

$\Rightarrow \{(x,y) : x \in A \cup C, y \in B \cup D\}$.

Therefore, once again, these sets must be equivalent to each other.

Now we have $(A \times B) \setminus (C \times D)$:

$\Rightarrow \{(a,b) : a \in A, b \in B\} \setminus \{(c,d) : c \in C, d \in D\}$

$\Rightarrow \{(x,y) : (x,y) \notin \{(c,d) : c \in C, d \in D\}\}$.

Thus, each ordered pair (x,y) is treated as an object, whereas $(A \setminus C) \times (B \setminus D)$ evaluates individual (x,y) such that $x \in A \setminus C$ and $y \in B \setminus D$. Therefore, $(A \times B) \setminus (C \times D) \neq (A \setminus C) \times (B \setminus D)$.

7. We first show that by A3.11, there must be a function $h \in (X \times Y)^Z$ such that each input $z \in Z$ maps to the domain of $\pi_{X \times Y \to X}$. Therefore, the codomain of h is equal to the domain of this map, whereby a single $c \in X \times Y$ has $c \mapsto x \ \forall x \in X$. This thus creates the range of f, and each $z \in Z$ is mapped to an $x \in X$. Since there is a bijective correspondence, $\pi_{X \times Y \to X} \circ h = f$ and equivalently $\pi_{X \times Y \to Y} \circ h = g$ for some $h \in (X \times Y)^Z$ such that each pairing is equivalent to its pairing in $X \times Y$.

9. We have $(\bigcup_{\alpha \in I} A_\alpha) \cap (\bigcup_{\beta \in J} B_\beta) = \bigcup_{(\alpha,\beta) \in I \times J} (A_\alpha \cap B_\beta)$.

Then we expand:

$(\bigcup_{\alpha \in I} A_\alpha) \cap (\bigcup_{\beta \in J} B_\beta)$

$\Rightarrow (\{x : x \in A_\alpha \text{ for some } \alpha \in I\}) \cap (\{y : y \in B_\beta \text{ for some } \beta \in J\})$.

6. $A \times B \subseteq C \times D$

$\Rightarrow \{(a,b) : a \in A, b \in B\} \subseteq \{(c,d) : c \in C, d \in D\}$

We thus use A3.6 to separate the ordered pair:

$\Rightarrow a \in C \ \forall a \in A, b \in D \ \forall b \in B$; thus by D3.1.14 this implies $A \subseteq C$ and $B \subseteq D$.

We now show $A \times B = C \times D$ iff $A = C$ and $B = D$. Suppose $A \times B \neq C \times D$. Then there is an element $x \in A \times B$ that $x \notin C \times D$. Thus, this is only true if for each $x = (a,b)$, $a \in A \not\Rightarrow a \notin C$ or $b \in B \not\Rightarrow b \in D$. Thus, for the opposite to be true, $a \in A \Rightarrow a \in C$, $b \in B \Rightarrow b \in D \ \forall a \in A, \forall b \in B$ and vice versa. These sets must therefore have the same pairs for all elements, and $A \times B = C \times D$. If either A or B or both are empty, $A \times B \subseteq C \times D$ still holds. At least one additional set: C, D, or both, must be empty for the relation to remain true. Similar logic applies to the second statement.

8. We have $\prod_{i=1}^{n} X_i = \emptyset$ iff at least one $X \in \{X_1, \dots, X_n\}$ is empty. Suppose, for contradiction, that one X_i was empty and $\prod_{i=1}^{n} X_i = \emptyset$. Then, for each $(x_1, \dots, x_n) \in \prod_{i=1}^{n} X_i$, there must be some $x_i \in X_i \ \forall i \in [1,n]$. However, by A3.3, one X_i is defined as having no x_i such that $x_i \in X_i$. Thus, $x_i \in Y_i \ \forall i \in [1,n]$ is never true and no $(x_1, \dots, x_n)$ satisfy these criterion. Thus, $\prod_{i=1}^{n} X_i = \{\} = \emptyset$. Having no empty sets, empty set leads to a contradiction whereby $x_i \notin Y_i \ \forall i \in [1,n]$. Thus, $(x_1, \dots, x_n)$ is still always false.

↑ Induction!

The intersection has some $z \in \{x : \dots\}$ iff $z \in A_\alpha$ and $z \in B_\beta$ for some $(\alpha, \beta) \in I \times J$. By A3.3, we replace this property with the intersection $(A_\alpha \cap B_\beta)$ such that $(\bigcup_{\alpha \in I} A_\alpha) \cap (\bigcup_{\beta \in J} B_\beta) = \bigcup_{(\alpha,\beta) \in I \times J} (A_\alpha \cap B_\beta)$.

If we interchanged the union $\cup$ symbols with intersection, we'd have $(\bigcap_{\alpha\in I} A_\alpha)\cup(\bigcap_{\beta\in J} B_\beta) = \bigcap_{(\alpha,\beta)\in I\times J}(A_\alpha \cup B_\beta)$.

Thus, the lefthand expression contains elements x, by definition, such that $x\in A_\alpha\ \forall \alpha\in I$, or $x\in B_\beta\ \forall \beta\in J$, or both. Thus, the righthand expression contains elements x such that $x\in A_\alpha\cup B_\beta\ \forall(\alpha,\beta)\in I\times J$. Informally, there may be some $x\in A_\alpha\ \forall\alpha\in I$ that $x\notin B_\beta\ \forall\beta\in J$ in the lefthand expression, whereby $x\notin A_\alpha\cup B_\beta\ \forall(\alpha,\beta)\in I\times J$. Thus these two statements are not equivalent.

10. i. Suppose $f=\tilde{f}$ but these two functions do not share the same graph. Thus $\{(x,f(x)) : x\in X\}\neq\{(x,\tilde{f}(x)) : x\in X\}$. Thus, since $x=x\ \forall x\in X$, then this implies $f(x)\neq\tilde{f}(x)$ for some $x\in X$ for these two sets to be inequivalent. Then, the same input $x\in X$ maps to different outputs, and therefore D.3.3.8 asserts $f\neq\tilde{f}$. We thus have a contradiction, and the graphs must be equivalent to eachother.

ii. We use replacement, whereby for each $x\in X$, $\{y\in Y : (x,y)\in G\}$ must also be true. We can thus have all functions $f\in Y^X$ that map $x\mapsto y$ iff $(x,y)\in G$. Since there is injectivity, having one f map multiple $x\in X$ to one $y\in Y$ invalidates the property. There is thus only one function whereby $\{f(x)=y : (x,y)\in G\}$ holds.

iii. Regardless of domain X, Y, or property $P(x,y)$, this definition has an input $x\in X$ to an output $y\in Y$. Based on $P(x,y) := (x,y)\in G$, there will always be one $y\in Y$ for each $x\in X$ such that $P(x,y)$ is true. Thus, by 4.3.7 we have $P(x,f(x))$ being true and for any $x\in X$ and $y\in Y$ we have: $y=f(x) \iff P(x,y)$ is true. We prove this as G is a subset of X,Y, whereby a pairing with any $x\in X$ is defined to have an ordered pair (x,y). We therefore defined a function; as G remains always contained within $X\times Y$.

11. We use $\exists$ 3.4.10 to construct the set $\{G : G\subseteq X\times Y\}$. Then, by A3.6, we have a set containing only the elements of $\{G : G\subseteq X\times Y\}$ whereby $P(x) := (G$ follows the vertical line test$)$ is true. Thus, for each $G\in\{G : G\subseteq X\times Y$ and $P(x)\}$, there must be a function $f: X\to Y$ whose graph is equivalent to G. We thus construct this set, of all f such that $f: X\to Y$ thus some $G\in\{G : G\subseteq X\times Y$ and $P(x)\}$ for which f defines the set $\{(x,f(x)) : x\in X\}$. We denote this set as Y^X, and thereby create the power set axiom.

$\{0,1,2,3,4\}\rightsquigarrow\{(n,f(n)),(n,f(n,f(n))),\ldots\}$

12. i. fix every natural number $N \in \mathbb{N}$, we have some function $a_N : \{n \in \mathbb{N} : n \leq N\} \to X$ such that $a_N(0) = c$ and $a_N(n++) = f(n, c_n(n))$ $\forall n \in \mathbb{N}$ such that $n < N$. Thus, since such $a_N(n++) \in \mathbb{N}$, then this function a_N is well-defined such that $a(0) = c$ and $a(n++) = f(n, a(n))$ $\forall n \in \mathbb{N}$.

ii. for each natural number $N \in \mathbb{N}$, there is some unique pair A_N, B_N of subsets of $\mathbb{N}$ that hold:

(a) $A_N \cap B_N = \emptyset$ $\Rightarrow$ each element $n \in \mathbb{N}$ is unique and therefore one can construct a set such that $A_N := \{n \in \mathbb{N} : n < N\}$ and $B_N := \{n \in \mathbb{N} : n \geq N\}$. Thus $A_N \cap B_N = \emptyset$ $\forall n \in \mathbb{N}$.

(b) $A_N \cup B_N = \mathbb{N}$. Thus, since all numbers $x \in \mathbb{N}$ must hold the following properties extensionally: $x < N$, $x = N$, $x > N$, having $A_N \cup B_N \Rightarrow \{n \in \mathbb{N} : n < N\} \cup \{n \in \mathbb{N} : n \geq N\} = \{n \in \mathbb{N} : n < N, n = N, n > N\}$ which therefore consists of $\mathbb{N}$.

(c) Since no element has $n < 0$, $0 \in A_N$ as the contrary would provide: $0 \in B_N \Rightarrow 0 \geq N$ for some $N \in \mathbb{N}$. If $n \neq 0$, then B_n is empty as no number has $n++ = 0$. Thus, $0 \in A$ always. since $0 < N$ is true for all $n \in \mathbb{N}$ provided $n \neq 0$.

(d) $N++ \in B_N$. Since $N \in B_N$ given that $N \geq N \Rightarrow N = N$. Then, $N++$ has, by definition, $N++ > N$. Thus $N++ \geq N$ is also true, and $N++ \in B_N$.

(e) Replace all $N = n$, $N++ = n++$. With A.3.7, this proof is self-evident from section d.

(f) If $n \in A_N$ and $n \neq N$ then $n++ \in A_N$. We modify $A_N := \{n \in \mathbb{N} : n \leq N\}$ such that whenever $n \in A_N$ and $n \neq N$, $n++$ has either $n++ < N$, or $n++ = N$. In both cases, $n++ \leq N$ and $n++ \in A_N$.

Thus, we substitute $\{n \in \mathbb{N} : n \leq N\}$ with A_N, and $\{n \in \mathbb{N} : n > N\}$ with B_N. Then we let $c \in \mathbb{N}$, such that some function $f : \mathbb{N} \times X \to X$ has an be defined as $c_1(0) = c$. Thus, $n++$ will also be contained within f, such that $a(n++) = f(n, a(n))$ $\forall n \in \mathbb{N}$.

13. We have $a(0) = c$ whereby $c = 0'$. Thus, using our examples above, $a(n) = n'$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$. This function is thus unique, and no other $a_1(x) = n'$ provided $x \neq n$. We therefore have a bijection. Now, define some property $n++ := n'++$, whereby bijectivity is preserved: $a(n++) = n'++$ iff $n++ = n'++$. Thus, since domains, codomains and all properties thereof equate, $\mathbb{N} = \mathbb{N}'$. There is thus only one set of all natural numbers. QED.

ZZZ.



3.6 Cardinality of Sets

1. Following the reflexivity axiom, we assert $X=X$. Thus, let $P(x)$ denote some arbitrary cardinality property. Regardless of how $P(x)$ is defined, we hold that $P(X)=P(X)$ by substitution. We also prove that a bijection exists. X, by definition, contains all elements such that $x\in X$ is true for some arbitrary X. There exists a function mapping every $y\in Y$ by exactly one $x\in X$ such that $f(x)=y$. In particular, since $X=X$, every $y\in X$ has one $x\in X$ thereby $y=x$. Now, suppose for contradiction that X and Y having equal cardinality does not imply having Y equal cardinality with X. Then, there is no bijective function $f: Y\to X$. However, some bijective function $g: X\to Y$, by definition, exists. By E.3.3.6, the inverse $g^{-1}: Y\to X$ must exist and therefore also be a bijection. We therefore have a contradiction, and our proof is done. Now, define $f: X\to Y$ and $g: Y\to Z$ as bijective functions. Then the function $g\circ f: X\to Z$ must also be a bijection by E.3.3.7. Thus, there is a function $h=g\circ f$ mapping X to Z bijectively, therefore X and Z have the same cardinality.

2. Let $C:=\{i\in\mathbb{N}: 1\le i\le n\}$ such that X has the same cardinality with C by D.3.6.5. Then, at $n=0$, $C:=\{i\in\mathbb{N}: 1\le n\le 0\}$. We then induct on i to show no element $x\in C$. At $n=0$, $C:=\{i\in\mathbb{N}: 1\le 0\le 0\}$ which is absurd as $1\le 0\le 0$ implies $1=0$ or $1<0$; axiomatically these are false. Now suppose we defined this set by i, then at i^{++} we have $C:=\{i\in\mathbb{N}: 1\le i^{++}\le 0\}$. Since no $i^{++}\neq 0$ and no i^{++} is less than zero, C has only $i\in\mathbb{N}$ have $i\notin C$. We close the induction and state that any element x has $x\notin C$. This, by A.3.3, creates the empty set.

3. We induct on n. Let $n=1$, then $C:=\{i\in\mathbb{N}: 1\le i\le n\}$ has one element whereby this is true, namely n itself. For simplicity, let $f: C\to\mathbb{N}$ map $1\mapsto 1$ in this scenario. Then, $n=M$ has all $i\in C$ have $f(i)\le M$. Now suppose we further proved this with n. We prove n^{++}. $C:=\{i\in\mathbb{N}: 1\le i\le n^{++}\}$ has, by definition, n^{++} elements. Then map each $i\in C$ such that $f: C\to\mathbb{N}$ has $i\mapsto i$. Then, $n^{++}=M^{++}$ has all $i\in C$ be, by definition, at most n^{++}. $f(n^{++})=n^{++}$, and so $f(n^{++})\le M^{++}$. We therefore close the induction. For contradiction, suppose $\mathbb{N}$ is finite. Because $\mathbb{N}\subseteq\mathbb{N}$, this implies $\mathbb{N}$ is bounded. Thus, there exists some M such that in $C:=\{i\in\mathbb{N}: 1\le i\le n\}$ for an arbitrary n such that $f(i)=i\le M\ \forall i\in\mathbb{N}$. However, by definition, since $M\in\mathbb{N}$, $M^{++}\in\mathbb{N}$ as well. Thus, $f(M^{++})=M^{++}\le M$ is absurd, and we have a contradiction.

4. (a) Finity implies there is an $n\in\mathbb{N}$ such that the cardinality $\#(X\cup\{x\})=n$. We show that $f=\{x\}\to\{1\}$ has a bijective correspondence whereby $\{1\}=\{i\in\mathbb{N}: 1\le i\le n\}$ for $n=1$. Thus $\#(\{x\})=1$, and $\{x\}$ is finite. Thus, $\#(X)+\#(\{x\})=M$ for some $M\in\mathbb{N}$. Thus, $X\cup\{x\}$ is finite and $\#(X\cup\{x\})=\#(X)+1$.

(b) There may be elements $x \in X$ that also $x \in Y$. Thus, the set $X \cup Y$ may contain extraneous elements in both X and Y whose cardinality is, by definition, one. Thus, it results in having two instances of the same element count as one. Therefore $\#(X \cup Y) \leq \#(X) + \#(Y)$. Disjoint sets guarantees adding unique elements, and therefore $\#(X) + \#(Y) = \#(X \cup Y)$ as no element $x \in X \cup Y$ is in both X and Y, therefore there is a bijection $f: X \cup Y \to X$ and $g: X \cup Y \to Y$ such that the statement above is true.

(c) $Y \subseteq X \Rightarrow \#(Y) \leq \#(X)$. Therefore, if X is finite and $\#(X) = n$ for some $n \in \mathbb{N}$, then $\#(Y) \leq n$ and is therefore bounded. As such, Y is finite. If we have $Y \subsetneq X$, then $\#(Y) < \#(X)$ as there will always be one element $x \in X$ such that $x \notin Y$. Thus, there is no perfect bijection $f: X \to Y$, and therefore X and Y must have different cardinalities.

(d) $f: X \to Y$ produces $f(X)$, whereby it is finite given that, at most, all elements $f(x)$ are seen once. Define $\#(Y) \leq \#(X)$, wherefrom substitution has $\#(f(x)) \leq \#(X)$. Then, iff X is injective, then $\#(f(x)) = \#(X)$ as Y can be replaced by the set $\{i \in \mathbb{N} : 1 \leq i \leq n\}$ whereby $f: X \to Y$ for $n \geq \#(X)$ always gives a unique element $f(x) = y$ for some $x \in X$. Thus injectivity equates to equal cardinality.

(e) $X \times Y$, by definition, has all ordered pairs with $x \in X$ and $y \in Y$. Thus, $X \times Y$ is finite as it is bounded by its products X and Y, which are both finite. Therefore, since $X \times Y$ contains all objects (x,y), we may separate as such: $\#\{(x_1,y_1), \ldots, (x_i,y_i)\} = Y$ and called all these Y_i $\forall i \in \mathbb{N}$. Though not yet defined, this is analogous to $\#(X) \times \#(Y)$.

(f) Y^X, as above is bounded by its constituents X and Y. Thus Y^X is finite as well and applying similar trivial logic, we arrive at the conclusion that $\#(Y^X) = \#(Y)^{\#(X)}$

5. We define $f: (A \times B) \to (B \times A)$ such that $f(x,y) = (y,x)$. Thus, each $(y,x) \in (B \times A)$ has exactly one $(x,y) \in (A \times B)$ for which $f(x,y) = (y,x)$ is true. Therefore f is a bijection and those two sets have equal cardinality. Since both $(A \times B)$ and $(B \times A)$ are equal, P.3 6.14 (c) asserts that $\#(X \times Y) = \#(Y \times X)$, whereby E.2.3.1 shows that $n \times m = m \times n$. One can adapt similar logic here.

6. Using similar logic above, there exists a function $f: (A^B)^C \to A^{B \times C}$ whereby $f((a^b)^c) = a^{b \times c}$. Thus, since this function is bijective, we arrive at a similar conclusion above whereby $(a^b)^c = a^{bc}$ for $a,b,c \in \mathbb{N}$. Therefore a

the trivial analogue $a^b \times a^c = a^{bc}$ is also true.

7. In the case whereby $\#(A) = \#(B)$, this implies the bijection $f: A \to B$ exists, whereby injectivity is already true. Without surjectivity, there may be an element $x \in B$ such that $x \notin A$. Thus, $\#(A \cup \{x\}) = \#(A) + 1$, whereby we induct on the cardinality of $\{x\}$. Suppose $\#\{x_1, \dots, x_n\} = n$ hence all $x_i \in B$ but $x_i \notin A$. Then $\#\{x_1, \dots, x_{n+1}\} = n+1$ thus $\#(A \cup \{x_1, \dots, x_{n+1}\}) = \#(A) + n+1$, which clearly means $\#(A) \leq \#(A \cup \{x_1, \dots, x_{n+1}\}) \Rightarrow \#(A) \leq \#(B)$.

8. We have $\#(A) \leq \#(B)$. Thus, $\#(B) \geq \#(A)$. Thus, by definition, surjectivity implies that for a set A with cardinality n, there is a set x that maps some $x \in X$ to each $y \in A$ through $f(x) = y$. Thus, $\#(B) \geq \#(A)$ must hold for surjectivity as A is non-empty, meaning that a function exists mapping elements of A to elements of B.

9. Since both A and B are finite sets, then by P.3.6.14 (b) $A \cup B$ is finite. By definition, $A \cap B \subseteq A \cup B$ and therefore by P.3.6.14 (c) $A \cap B$ is finite. Also, $\#(A) + \#(B) \geq \#(A \cup B)$ by P.3.6.14 (b). Then, $\#(A \cup B) \leq \#(A \cup B) + \#(A \cap B)$. Then, let $n = \#(A \cap B)$ such that $n \in \mathbb{N}$. Then, $\#(A \cup B) \leq \#(A \cup B) + n$, which by definition, must be true.

10. Suppose, for contradiction, that we take the Legion's contrary and let $\#(A_i) = 1 \quad \forall i \in \{1, \dots, n\}$. Then, $\bigcup_{i \in \{1,\dots,n\}} A_i = A_1 \cup \dots \cup A_n$. Hence, in the logical extreme we have $\#(\bigcup_{i \in \{1,\dots,n\}} A_i) = \#(A_1) + \dots + \#(A_n)$ by P.3.6.14 (b). We therefore have $\#(\bigcup_{i \in \{1,\dots,n\}} A_i) \in n$. Which, must be a contradiction. Thus, there needs to be some $i \in \{1, \dots, n\}$ such that $\#(A_i) \geq 2$.

11. $\#(E) = 2$ implies E has two elements $E := \{x_1, x_2\}$ for $x_1, x_2 \in X$. Thus, by definition, $f(x_1) \neq f(x_2)$. Therefore, if we let $C := \{f(x_1)\}$ and $\{f(x_2)\}$, it can be shown $f(x_2) \notin C$. Thus, $C + \{f(x_2)\} = \#(C) + 1 = 1 + 1 = 2$. Thus, injectivity must imply two-to-two properties. The opposite holds true, whereby $\#(E) = 2 \Rightarrow \#(f(E)) = 2 \quad \forall E \subseteq X$ means no element $x_1 \in X$ and $x_2 \in X$ has $f(x_1) = f(x_2)$. Thus it must be injective.

12. i. Suppose we proved S_n is finite and $\#(S_{n+1}) = (n+1) \times \#(S_n)$. We partition S_{n+1} into $n+1$ subsets. Thus, this $\phi(n+1)$ has $\phi: \{i \in \mathbb{N} : 1 \leq i \leq n+1\} \to \{i \in \mathbb{N} : 1 \leq i \leq n+1\}$ such that S_{n+1} assigns to $n+1$. Then, S_{n+1} must also be finite and therefore the resultant term has $\#(S_{n+1}) = (n+1) \times \#(S_n)$.

ii. Let the factorial of some $n \in \mathbb{N}$ be defined such that $0! := 1$ and $(n+1)! := (n+1) \times n!$ Then, $\#(S_n) = n!$ $\forall n \in \mathbb{N}$ as inductively $\#(S_{n+1}) = n+1 \times \#(S_n)$. Thus, $\#(S_{n+1}) = (n+1)! \Rightarrow \#(S_{n+1}) = (n+1) \times n!$ It has been shown, then, that hence $\#(S_n) = n! \Rightarrow \#(S_{n+1}) = (n+1) \times \#(S_n)$. Which, as proved, must be true.